

# Generation Interconnection Combined Feasibility / System Impact Study Report

## For Queue Project Q20-1532

Speed 138 kV  
December 27, 2021

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1532. The Interconnection Customer proposes a 79.0 MW wind facility interconnecting at the Speed 138 kV substation in the Kentucky area. The request was received on July 26, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 495 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,891,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility               | Contingency                             | Loading (%) | Rating (MVA) | DFAX (%) |
|----------------------------------|-----------------------------------------|-------------|--------------|----------|
| GAGA-BARNHAM 138kV               | SPEED-SILVER BAY POWER 138kV            | 167.7       | 190          | 100.0    |
| SPEED-GAGA 138kV L               | SPEED-SILVER BAY POWER 138kV            | 161.9       | 209          | 100.0    |
| SILVER BAY POWER-SPEED 138kV     | SPEED-GAGA 138kV L                      | 159.5       | 212          | 100.0    |
| NAROD-BARNHAM 138kV              | SILVER BAY POWER-SPEED 138kV line       | 156.4       | 191          | 100.0    |
| SANTA MONICA-HAMBRICK 138kV Line | SPEED-SILVER BAY POWER 138kV line       | 127.7       | 218          | 100.0    |
| BROCKPORT-NAROD 138kV            | SPEED-SILVER BAY POWER 138kV            | 119.9       | 221          | 100.0    |
| 56TH ST-SANTA MONICA 138kV       | SILVER BAY POWER-SPEED SUBSTATION 138kV | 113.9       | 228          | 100.0    |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility           | Required Upgrade                            | Cost Estimate | Allocation to Project |
|--------------------|---------------------------------------------|---------------|-----------------------|
| GAGA-BARNHAM 138kV | Rebuild line with higher-capacity conductor | \$2,384,000   | 30% (\$706,000)       |

| Facility                          | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-----------------------------------|---------------------------------------------|---------------|-----------------------|
| SPEED-GAGA 138kV L                | Rebuild line with higher-capacity conductor | \$3,542,000   | 30% (\$1,048,000)     |
| SILVER BAY POWER-SPEED 138kV Line | Rebuild line with higher-capacity conductor | \$26,599,000  | 30% (\$7,873,000)     |
| BARNHAM-NAROD 138kV               | Rebuild line with higher-capacity conductor | \$12,507,000  | 30% (\$3,702,000)     |
| HAMBRICK-SANTA MONICA 138kV line  | Rebuild line with higher-capacity conductor | \$4,279,000   | 30% (\$1,267,000)     |
| BROCKPORT-NAROD 138kV             | Rebuild line with higher-capacity conductor | \$2,510,000   | 30% (\$743,000)       |
| SANTA MONICA-56TH ST 138kV L      | Rebuild line with higher-capacity conductor | \$1,865,000   | 30% (\$552,000)       |
|                                   | Total allocated to project                  |               | \$15,891,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.